

****ICSE CLASS 10 CHEMISTRY — COMPLETE REVISION NOTES****

****CHAPTER 1: PERIODIC TABLE****

****YouTube Lecture:**** <https://youtu.be/lePO2EmTQkQ?si=smF2po7LIKQuls71>

****1. Historical Development of the Periodic Table****

****A. Dobereiner's Triads****

Dobereiner arranged elements into groups of three called triads. In each triad, the atomic mass of the middle element was approximately the average of the atomic masses of the other two elements.

Example: Lithium (Li), Sodium (Na), Potassium (K).

Limitation: Dobereiner could identify only a few triads.

****B. Newlands' Law of Octaves****

Newlands arranged elements in increasing order of atomic masses. He found that the properties of every eighth element were similar to the properties of the first element, just like the eighth note in music.

Limitation: It worked well only for elements up to calcium. Later, new elements did not fit into this pattern.

****C. Mendeleev's Periodic Law****

Mendeleev stated that the physical and chemical properties of elements are a periodic function of their atomic masses.

Achievements: He left gaps for undiscovered elements and predicted their properties (e.g., eka-aluminium for gallium, eka-silicon for germanium).

Limitations: Position of hydrogen was not justified. Isotopes posed a problem. Anomalous pairs like cobalt (higher mass) placed before nickel (lower mass) could not be explained.

****2. Modern Periodic Law****

The modern periodic law states, "The physical and chemical properties of elements are a periodic function of their atomic numbers." This law was proposed by Henry Moseley. The law resolved the anomalies of Mendeleev's table because atomic number is a more fundamental property than atomic mass.

****3. The Modern Periodic Table****

The Modern Periodic Table consists of 18 vertical columns called groups and 7 horizontal rows called periods.

****A. Groups (1 to 18)****

Group 1: Alkali Metals (Li, Na, K, Rb, Cs, Fr)

Group 2: Alkaline Earth Metals (Be, Mg, Ca, Sr, Ba, Ra)

Group 17: Halogens (F, Cl, Br, I, At)

Group 18: Zero Group or Noble Gases (He, Ne, Ar, Kr, Xe, Rn)

****B. Periods (1 to 7)****

Period 1: Shortest period, contains only 2 elements (H and He).

Period 2 and 3: Short periods, contain 8 elements each.

Period 4 and 5: Long periods, contain 18 elements each.

Period 6: Longest period, contains 32 elements.

Period 7: Incomplete period.

****4. Periodic Properties and Their Variations (Across a Period and Down a Group)****

****A. Atomic Size (Atomic Radius)****

Across a Period (Left to Right): Atomic size decreases. Reason: Nuclear charge increases, pulling electrons closer to the nucleus.

Down a Group (Top to Bottom): Atomic size increases. Reason: New shells are added, increasing the distance between the nucleus and the outermost electrons.

****B. Metallic Character****

Across a Period (Left to Right): Metallic character decreases. Elements become less metallic and more non-metallic.

Down a Group (Top to Bottom): Metallic character increases.

****C. Non-Metallic Character****

Across a Period (Left to Right): Non-metallic character increases.

Down a Group (Top to Bottom): Non-metallic character decreases.

****D. Ionisation Potential (Ionisation Energy)****

Definition: The minimum energy required to remove the most loosely bound electron from an isolated neutral gaseous atom.

Across a Period (Left to Right): Increases. Reason: Atomic size decreases, nuclear charge increases, so electrons are held more tightly.

Down a Group (Top to Bottom): Decreases. Reason: Atomic size increases, outermost electron is farther from the nucleus and easier to remove.

****E. Electron Affinity****

Definition: The amount of energy released when a neutral isolated gaseous atom gains an electron to form a gaseous anion.

Across a Period (Left to Right): Increases (becomes more negative).

Down a Group (Top to Bottom): Decreases (becomes less negative).

****F. Electronegativity****

Definition: The tendency of an atom to attract shared electrons towards itself in a covalent bond.

Across a Period (Left to Right): Increases.

Down a Group (Top to Bottom): Decreases.

****5. Periodicity****

Periodicity is the repetition of similar properties of elements after a regular interval when elements are arranged in increasing order of their atomic numbers. This is because similar electronic configuration repeats after a certain interval. Elements in the same group have the same number of valence electrons, hence they show similar chemical properties.

****IMPORTANT QUESTIONS: PERIODIC TABLE****

****Q1.**** State Dobereiner's Law of Triads. Give one example of a triad. What was its limitation?

****Q2.**** State Newlands' Law of Octaves. Why was it discarded?

****Q3.**** State Mendeleev's Periodic Law. Mention two achievements and two limitations of Mendeleev's Periodic Table.

****Q4.**** State the Modern Periodic Law. Who proposed it?

****Q5.**** How many groups and periods are there in the Modern Periodic Table?

****Q6.**** Name the following groups:

- (a) Group 1
- (b) Group 17
- (c) Group 18

****Q7.**** Define the following periodic properties and state how they vary across a period and down a group:

- (a) Atomic size
- (b) Ionisation potential
- (c) Electron affinity
- (d) Electronegativity

****Q8.**** Explain why atomic size decreases across a period but increases down a group.

****Q9.**** With reference to Group 1 (Alkali metals), name the elements present in this group.

****Q10.**** What is periodicity? Why do elements in the same group show similar chemical properties?

****CHAPTER 2: CHEMICAL BONDING****

****YouTube Lecture:**** <https://youtu.be/oFPPJstX7N0?si=BhF9E205BSDdhPgg>

****1. Introduction to Chemical Bonding****

Chemical bonding is the force of attraction that holds atoms together in a molecule or compound. Atoms combine to achieve a stable electronic configuration, usually an octet (8 electrons) or duplet (2 electrons for hydrogen) in their outermost shell.

****2. Electrovalent Bond (Ionic Bond)****

An electrovalent bond is formed by the complete transfer of one or more electrons from a metal atom to a non-metal atom. This results in the formation of positively charged cations and negatively charged anions. The electrostatic force of attraction between these oppositely charged ions holds them together.

****A. Electron Dot Structures of Electrovalent Compounds****

****NaCl (Sodium Chloride):**** Sodium (2,8,1) loses one electron to form Na^+ . Chlorine (2,8,7) gains one electron to form Cl^- . The oppositely charged ions attract each other.

****MgCl₂ (Magnesium Chloride):**** Magnesium (2,8,2) loses two electrons to form Mg^{2+} . Two chlorine atoms each gain one electron to form two Cl^- ions.

****CaO (Calcium Oxide):**** Calcium (2,8,8,2) loses two electrons to form Ca^{2+} . Oxygen (2,6) gains two electrons to form O^{2-} .

****B. Characteristic Properties of Electrovalent Compounds****

Physical State: Usually crystalline solids.

Melting and Boiling Points: High, due to strong electrostatic forces of attraction between ions.

Electrical Conductivity: Do not conduct electricity in solid state but conduct in molten state or aqueous solution because ions become free to move.

Solubility: Generally soluble in polar solvents like water but insoluble in organic solvents.

****3. Covalent Bond****

A covalent bond is formed by the mutual sharing of electrons between two non-metal atoms. The shared electrons belong to both atoms and help each atom achieve a stable electronic configuration.

****A. Electron Dot Structures of Covalent Molecules****

Non-Polar Covalent Compounds (Equal sharing of electrons)

H₂ (Hydrogen): Two hydrogen atoms share one electron each to achieve a duplet.
Cl₂ (Chlorine): Two chlorine atoms share one electron each to achieve an octet.
O₂ (Oxygen): Two oxygen atoms share two electrons each, forming a double bond.
N₂ (Nitrogen): Two nitrogen atoms share three electrons each, forming a triple bond.
CH₄ (Methane): Carbon shares four electrons with four hydrogen atoms, forming four single covalent bonds.
CCl₄ (Carbon Tetrachloride): Carbon shares four electrons with four chlorine atoms.

Polar Covalent Compounds (Unequal sharing due to difference in electronegativity)
HCl (Hydrogen Chloride): Chlorine is more electronegative, so it attracts the shared electron pair more strongly, creating partial charges.
H₂O (Water): Oxygen is more electronegative than hydrogen, so O-H bonds are polar.
NH₃ (Ammonia): Nitrogen is more electronegative than hydrogen.

B. Characteristic Properties of Covalent Compounds

Physical State: Can be gases, liquids or soft solids.
Melting and Boiling Points: Low, due to weak intermolecular forces.
Electrical Conductivity: Do not conduct electricity in any state (no free ions).
Solubility: Generally insoluble in water but soluble in organic solvents.

4. Coordinate Bond (Dative Bond)

A coordinate bond is a covalent bond in which the shared electron pair comes from only one of the combining atoms. The atom that donates the electron pair is called the donor, and the atom that accepts it is the acceptor.

Key Concept: Lone Pair

A lone pair is a pair of valence electrons that is not shared with another atom and remains on the atom.

Formation of Hydronium Ion (H₃O⁺):

Water molecule (H₂O) has two lone pairs on the oxygen atom. It donates one of its lone pairs to a hydrogen ion (H⁺), which has no electrons, forming a coordinate bond. The resulting species is the hydronium ion (H₃O⁺).

Formation of Ammonium Ion (NH₄⁺):

Ammonia molecule (NH₃) has one lone pair on the nitrogen atom. It donates this lone pair to a hydrogen ion (H⁺), forming a coordinate bond and producing the ammonium ion (NH₄⁺).

5. Comparison of Electrovalent and Covalent Compounds

Property	Electrovalent Compounds	Covalent Compounds
Formation	Transfer of electrons	Sharing of electrons
Constituent Particles	Ions	Molecules

Physical State	Usually crystalline solids	Gases, liquids or soft solids
Melting and Boiling Points	High	Low
Electrical Conductivity	Conduct in molten/aqueous state	Do not conduct
Solubility	Mostly soluble in water	Mostly insoluble in water

****IMPORTANT QUESTIONS: CHEMICAL BONDING****

****Q1.**** What is the Octet Rule? Explain why atoms combine with each other.

****Q2.**** Draw the electron dot structures of the following:

- (a) Sodium Chloride (NaCl)
- (b) Magnesium Chloride (MgCl₂)
- (c) Calcium Oxide (CaO)

****Q3.**** Give any three characteristic properties of electrovalent compounds.

****Q4.**** Draw the electron dot structures of the following covalent molecules:

- (a) H₂
- (b) O₂
- (c) N₂
- (d) CH₄
- (e) CCl₄

****Q5.**** Differentiate between polar and non-polar covalent compounds with an example of each.

****Q6.**** State any three differences between electrovalent and covalent compounds.

****Q7.**** What is a coordinate bond? Explain its formation with reference to the hydronium ion and ammonium ion using electron dot diagrams.

****Q8.**** Why do electrovalent compounds conduct electricity in molten state but not in solid state?

****Q9.**** Why do covalent compounds have low melting and boiling points?

****Q10.**** Define the term 'lone pair'. Which atoms possess lone pairs in H₂O and NH₃?

****CHAPTER 3: ACIDS, BASES AND SALTS****

****YouTube Lecture:**** https://youtu.be/6gZsd4SG6hU?si=Hc_Bs2Dq3SHKIKMy

****1. Definitions and Characteristic Properties****

****A. Acids****

Acids are substances that dissociate in aqueous solution to produce hydrogen ions (H^+) or hydronium ions (H_3O^+). $H^+ + H_2O \rightarrow H_3O^+$

Properties: Sour taste. Turn blue litmus red. Conduct electricity in aqueous solution.

Examples: Hydrochloric acid (HCl), Sulphuric acid (H_2SO_4), Nitric acid (HNO_3).

****B. Bases (Alkalis)****

Bases are substances that dissociate in aqueous solution to produce hydroxyl ions (OH^-). A water soluble base is called an alkali.

Properties: Bitter taste, soapy touch. Turn red litmus blue. Conduct electricity in aqueous solution.

Examples: Sodium hydroxide (NaOH), Potassium hydroxide (KOH), Calcium hydroxide $Ca(OH)_2$.

****C. Salts****

A salt is formed by the partial or complete replacement of the hydrogen ion (H^+) of an acid by a metal ion or ammonium ion (NH_4^+).

Example: $HCl + NaOH \rightarrow NaCl + H_2O$

****2. Ions Present in Acids, Bases and Salts****

Acids produce H^+ (hydronium H_3O^+) ions as the only positive ions.

Bases produce OH^- (hydroxyl) ions as the only negative ions.

Salts produce metal or ammonium cations and acid radical anions.

****3. Acid-Base Indicators and pH Scale****

****Common Indicators:****

Blue Litmus: Turns red in acid.

Red Litmus: Turns blue in base.

Methyl Orange: Red in acid, yellow in base.

Phenolphthalein: Colourless in acid, pink in base.

****pH Scale:**** The pH scale ranges from 0 to 14. pH less than 7 is acidic, pH 7 is neutral, pH greater than 7 is basic. pH is tested using pH paper or universal indicator.

****4. Types of Salts****

****Normal Salt:**** Formed by complete replacement of H^+ of an acid by a metal ion. Example: NaCl, Na_2SO_4 , KNO_3 .

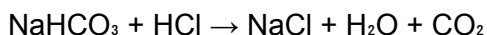
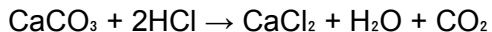
****Acid Salt:**** Formed by partial replacement of H^+ of an acid containing more than one replaceable hydrogen. Example: $NaHCO_3$, $NaHSO_4$.

****Basic Salt:**** Formed by partial replacement of OH^- of a base by an acid radical. Example: $Pb(OH)Cl$, $Mg(OH)Cl$.

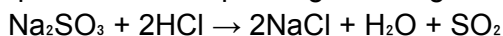
****Complex Salt:**** Contains a complex ion (central metal ion surrounded by ligands).
Example: Sodium argentocyanide $\text{Na}[\text{Ag}(\text{CN})_2]$.

****5. Action of Dilute Acids on Salts****

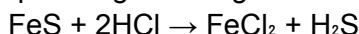
Dilute acids decompose carbonates and bicarbonates with effervescence of CO_2 .



Sulphites and bisulphites give SO_2 gas.

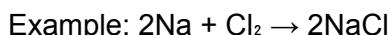


Sulphides give H_2S gas.



****6. Methods of Preparation of Normal Salts****

****Direct Combination:**** Heated metal + non-metal.



****Displacement:**** Active metal displaces less active metal from its salt solution.



****Precipitation (Double Decomposition):**** Mixing two soluble salts to form an insoluble salt.



****Neutralisation of Insoluble Base:**** Acid + Insoluble base (e.g., CuO , ZnO).



****Neutralisation of Alkali (Titration Method):**** Acid + Alkali.



****Action of Dilute Acid on Carbonate/Bicarbonate:****



****IMPORTANT QUESTIONS: ACIDS, BASES AND SALTS****

****Q1.**** Define acids and bases in terms of ions produced by them in aqueous solution.

****Q2.**** What are hydronium ions? Write an equation to show their formation.

****Q3.**** Explain the pH scale. How is pH paper used to determine the nature of a solution?

****Q4.**** What is a salt? Name four types of salts with one example each.

****Q5.**** Differentiate between a normal salt and an acid salt.

****Q6.**** Name the gas evolved when dilute hydrochloric acid is added to:

- (a) Zinc carbonate
- (b) Sodium sulphite
- (c) Iron sulphide

Write the balanced chemical equation for each reaction.

Q7. Describe any three methods for the preparation of normal salts with suitable examples and equations.

Q8. Why is lime water used as an indicator for carbon dioxide?

Q9. State the colours of the following indicators in acidic and basic medium:

- (a) Methyl Orange
- (b) Phenolphthalein

Q10. What is an alkali? Is every base an alkali? Explain.

CHAPTER 4: ANALYTICAL CHEMISTRY

YouTube Lecture: https://youtu.be/GNKb__9uwcM?si=NS980--9fyOVIRtn

1. Action of Sodium Hydroxide (NaOH) Solution on Salt Solutions

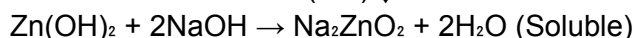
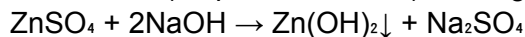
Sodium hydroxide reacts with salt solutions to form insoluble metal hydroxides (precipitates). The colour and solubility of the precipitate helps to identify the metal ion present.

Salts of Fe^{2+} (Ferrous salts): Dirty green precipitate of ferrous hydroxide $\text{Fe}(\text{OH})_2$. Insoluble in excess NaOH.

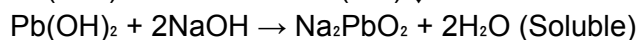
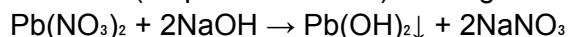
Salts of Fe^{3+} (Ferric salts): Reddish brown precipitate of ferric hydroxide $\text{Fe}(\text{OH})_3$. Insoluble in excess NaOH.

Salts of Cu^{2+} (Copper salts): Bluish (Pale blue) precipitate of copper hydroxide $\text{Cu}(\text{OH})_2$. Insoluble in excess NaOH. On heating, the precipitate turns black.

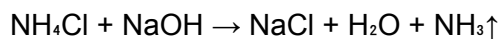
Salts of Zn^{2+} (Zinc salts): White gelatinous precipitate of zinc hydroxide $\text{Zn}(\text{OH})_2$. Soluble in excess NaOH (amphoteric nature) forming colourless sodium zincate.



Salts of Pb^{2+} (Lead salts): White curdy precipitate of lead hydroxide $\text{Pb}(\text{OH})_2$. Soluble in excess NaOH (amphoteric nature) forming colourless sodium plumbite.



Salts of NH_4^+ (Ammonium salts): On warming with NaOH, ammonia gas (NH_3) is evolved. Ammonia turns red litmus blue and gives white fumes with HCl.

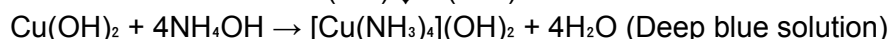
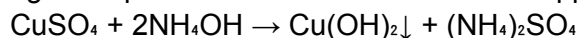


****2. Action of Ammonium Hydroxide (NH₄OH) Solution on Salt Solutions****

****Salts of Fe²⁺.**** Dirty green precipitate of Fe(OH)₂. Insoluble in excess NH₄OH.

****Salts of Fe³⁺.**** Reddish brown precipitate of Fe(OH)₃. Insoluble in excess NH₄OH.

****Salts of Cu²⁺.**** Bluish precipitate of copper hydroxide Cu(OH)₂. Soluble in excess NH₄OH forming a deep blue solution of tetraammine copper (II) complex.



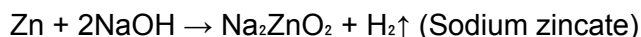
****Salts of Zn²⁺.**** White gelatinous precipitate of Zn(OH)₂. Soluble in excess NH₄OH forming a colourless solution of tetraammine zinc complex.

****Salts of Pb²⁺.**** White curdy precipitate of Pb(OH)₂. Insoluble in excess NH₄OH.

****3. Action of Alkalis (NaOH/KOH) on Certain Metals, Their Oxides and Hydroxides (Amphoteric Nature)****

Certain metals like Aluminium (Al), Zinc (Zn) and Lead (Pb), their oxides and hydroxides, react with strong alkalis (NaOH, KOH) to form soluble salts. They also react with acids. This property is called amphoteric nature.

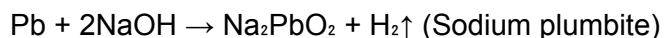
****With Zinc.****



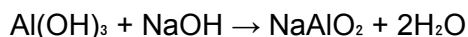
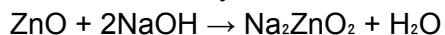
****With Aluminium.****



****With Lead.****



Their oxides and hydroxides also react similarly:



****IMPORTANT QUESTIONS: ANALYTICAL CHEMISTRY****

****Q1.**** State what is observed when sodium hydroxide solution is added drop by drop and then in excess to solutions of:

(a) Zinc sulphate

(b) Copper sulphate

(c) Ferrous sulphate

(d) Lead nitrate

Write balanced chemical equations.

****Q2.**** Explain the special action of ammonium hydroxide on copper sulphate solution. Write the relevant equations.

****Q3.**** On warming ammonium chloride with sodium hydroxide solution, a gas is evolved. Name the gas. Give two tests to identify the gas.

****Q4.**** What do you understand by the term "amphoteric nature"? Explain with reference to zinc, aluminium and lead. Give equations.

****Q5.**** Name the precipitate formed when sodium hydroxide is added to ferric chloride solution. State its colour.

****Q6.**** What is the colour of the following solutions?

- (a) Copper sulphate
- (b) Ferrous sulphate

****Q7.**** How will you distinguish between zinc ions and lead ions using ammonium hydroxide?

****Q8.**** Write the formula of sodium zincate and sodium aluminate.

****Q9.**** Why is ammonium hydroxide called a weak alkali?

****Q10.**** A white gelatinous precipitate soluble in excess NaOH is formed when a salt solution is treated with NaOH. Identify the metal ion.

****CHAPTER 5: MOLE CONCEPT AND STOICHIOMETRY****

****YouTube Lecture:**** <https://youtu.be/R9l0boD4taY?si=Q8xSitu4QRQbtgs9>

****1. Gay Lussac's Law of Combining Volumes****

When gases react together under the same conditions of temperature and pressure, they do so in simple whole number ratios by their volumes, and the volumes of the products (if gases) also bear a simple ratio to the volumes of the reactants.

Example: $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (1 volume : 3 volumes : 2 volumes)

****2. Avogadro's Law****

Equal volumes of all gases, under the same conditions of temperature and pressure, contain an equal number of molecules.

Molar Volume: At standard temperature and pressure (S.T.P.), one mole of any gas occupies 22.4 litres. The mass of 22.4 litres of any gas at S.T.P. is equal to its molar mass (molecular mass in grams).

****3. Relationship Between Kelvin Scale and Celsius Scale****

Temperature in Kelvin (K) = Temperature in Celsius ($^{\circ}\text{C}$) + 273

Standard Temperature and Pressure (S.T.P.): 0°C (273 K) and 760 mm of Hg (1 atm).

****4. Atomicity****

The atomicity of an element is the number of atoms present in one molecule of that element.

Monoatomic: He, Ne, Ar (atomicity 1)

Diatomic: H_2 , O_2 , N_2 , Cl_2 (atomicity 2)

Tetratomic: P_4 (atomicity 4)

****5. Vapour Density and Molecular Mass****

Vapour Density (V.D.) is the ratio of the mass of a certain volume of a gas to the mass of an equal volume of hydrogen under the same conditions of temperature and pressure.

Relationship: Molecular Mass = $2 \times$ Vapour Density

****6. Empirical Formula and Molecular Formula****

Empirical Formula: The simplest whole number ratio of atoms of different elements in a compound. It is deduced from the percentage composition.

Molecular Formula: The actual number of atoms of each element in a molecule.

Molecular Formula = $n \times$ Empirical Formula

$n = \text{Molecular Mass} / \text{Empirical Formula Mass}$

****7. The Mole Concept****

A mole is a quantity of a substance that contains 6.022×10^{23} (Avogadro's number) elementary particles (atoms, molecules, ions).

Gram Atomic Mass: The atomic mass of an element expressed in grams. It is the mass of one mole of atoms.

Gram Molecular Mass: The molecular mass of a substance expressed in grams. It is the mass of one mole of molecules.

Number of Moles = Given Mass / Molar Mass

Number of Moles = Given Volume at S.T.P. / 22.4 L

****8. Calculations Based on Chemical Equations****

A balanced chemical equation gives the mole ratio of reactants and products. From this, the masses and volumes of substances involved can be calculated.

****IMPORTANT QUESTIONS: MOLE CONCEPT AND STOICHIOMETRY****

****Q1.**** State Gay Lussac's Law of Combining Volumes. Give one example with equation.

****Q2.**** State Avogadro's Law. What is molar volume?

****Q3.**** Convert the following temperatures to Kelvin scale:

(a) 27°C

(b) -73°C

****Q4.**** Define vapour density. How is it related to molecular mass?

****Q5.**** A compound has the following percentage composition: Carbon = 40%, Hydrogen = 6.67%, Oxygen = 53.33%. Its molecular mass is 60. Find its empirical formula and molecular formula.

****Q6.**** Define the term "Mole". State Avogadro's number.

****Q7.**** Calculate the number of moles in:

(a) 8g of oxygen gas (O_2)

(b) 11.2 litres of carbon dioxide at S.T.P.

****Q8.**** What mass of sodium chloride is formed when 5.3g of sodium carbonate reacts completely with dilute hydrochloric acid?

(Na = 23, C = 12, O = 16, Cl = 35.5)

Equation: $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$

****Q9.**** Calculate the volume of oxygen required at S.T.P. for the complete combustion of 4 litres of methane.

Equation: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$

****Q10.**** Differentiate between empirical formula and molecular formula.

****CHAPTER 6: ELECTROLYSIS****

****YouTube Lecture:**** https://youtu.be/17_ukKwYE9U?si=qXT4sBi91bMOKu3t

****1. Key Definitions****

****Electrolysis:**** The process of decomposition of an electrolyte in its molten state or aqueous solution by the passage of direct electric current.

****Electrolyte:**** A compound that conducts electricity in molten state or aqueous solution and undergoes chemical decomposition.

****Non-Electrolyte:**** A compound that does not conduct electricity in any state.

****Electrodes:**** The conducting rods that dip into the electrolyte. The anode is connected to the positive terminal of the battery and the cathode is connected to the negative terminal.

****Anode:**** The positive electrode where oxidation (loss of electrons) occurs. Anions (negative ions) migrate towards the anode.

****Cathode:**** The negative electrode where reduction (gain of electrons) occurs. Cations (positive ions) migrate towards the cathode.

****2. Types of Substances Based on Composition****

****Substances containing molecules only:**** No free ions. These are non-electrolytes.
Example: Sugar, Glucose, Alcohol.

****Substances containing ions only:**** Ions held by strong electrostatic forces. These are solid electrolytes but conduct only in molten state. Example: NaCl solid.

****Substances containing both molecules and ions:**** These are strong electrolytes or weak electrolytes in aqueous solution. Example: H_2SO_4 , NaOH, HCl, NaCl solution.

****3. Selective Discharge of Ions****

In an electrolyte, there are usually more than one cation and more than one anion. Only one cation and one anion are discharged at the electrodes preferentially.

****Factors Affecting Selective Discharge:****

Position of the ion in the Electrochemical (Activity) Series: Lower the position in the series, easier it is to discharge at the cathode (cations). Higher the position in the series for anions, easier it is to discharge at the anode.

Concentration of the ion in the electrolyte.

Nature of the electrode.

****4. Examples of Electrolysis****

****A. Electrolysis of Molten Lead Bromide (Electrolyte: molten PbBr_2 , Electrodes: Graphite)****

Ions present: Pb^{2+} and 2Br^-

At Cathode: $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$ (Lead metal deposited, greyish)

At Anode: $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$ (Brown vapours of bromine)

Overall: $\text{PbBr}_2 \rightarrow \text{Pb} + \text{Br}_2$

****B. Electrolysis of Acidified Water (Electrolyte: water + few drops H_2SO_4 , Electrodes: Platinum)****

Ions present: H^+ , SO_4^{2-} , OH^-

At Cathode: $4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2$ (Hydrogen gas liberated, 2 volumes)

At Anode: $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ (Oxygen gas liberated, 1 volume)

Overall: $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

****C. Electrolysis of Aqueous Copper (II) Sulphate Using Copper Electrodes****

Ions present: Cu^{2+} , H^+ , SO_4^{2-} , OH^-

At Cathode: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ (Copper metal deposited on cathode, pinkish)

At Anode (Copper electrode is active): $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ (Anode dissolves, mass decreases)

Net effect: Copper is transferred from anode to cathode. Concentration of CuSO_4 remains constant.

****D. Electrolysis of Aqueous Copper (II) Sulphate Using Platinum (Inert) Electrodes****

At Cathode: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ (Copper deposited)

At Anode: $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$ (Oxygen gas evolved)

The blue colour of solution fades and solution becomes acidic due to H_2SO_4 .

****5. Applications of Electrolysis****

****A. Electroplating:**** Coating an inferior metal with a superior metal by electrolysis.

****Electroplating with Silver:**** Electrolyte is Sodium argentocyanide ($\text{Na}[\text{Ag}(\text{CN})_2]$). Anode is pure silver. Cathode is the article to be electroplated.

****Electroplating with Nickel:**** Electrolyte is Nickel sulphate solution. Anode is pure nickel. Cathode is the article to be electroplated.

****B. Electrorefining of Copper:**** Impure copper is made the anode, pure copper is the cathode, electrolyte is acidified copper sulphate solution. Pure copper is deposited on cathode and impurities fall as anode mud.

****IMPORTANT QUESTIONS: ELECTROLYSIS****

****Q1.**** Define the following terms:

- (a) Electrolysis
- (b) Electrolyte
- (c) Anode and Cathode

****Q2.**** Differentiate between strong electrolytes and weak electrolytes. Give one example of each.

****Q3.**** Explain the electrolysis of molten lead bromide with reactions at the electrodes.

****Q4.**** Draw a labelled diagram showing the electrolysis of acidified water. Give equations at the cathode and anode. Why is the volume of hydrogen collected double that of oxygen?

****Q5.**** Explain the electrolysis of copper sulphate solution using:

- (a) Copper electrodes
- (b) Platinum electrodes

Why does the colour of the solution fade in case (b) but not in case (a)?

****Q6.**** State the factors that influence the selective discharge of ions during electrolysis.

****Q7.**** What is electroplating? Name the electrolyte, anode and cathode used in silver electroplating.

****Q8.**** How is copper purified by electrolytic refining? Name the material used for anode and cathode.

****Q9.**** Differentiate between oxidation and reduction in terms of loss and gain of electrons.

****Q10.**** Explain why solid sodium chloride does not conduct electricity but its aqueous solution conducts.

****CHAPTER 7: METALLURGY****

****YouTube Lecture:**** <https://youtu.be/9DaA95KHpoA?si=vQO7Mn94QAnzncnC>

****1. Key Terms****

****Mineral:**** A naturally occurring substance containing one or more metals in the earth's crust.

****Ore:**** A mineral from which a metal can be extracted profitably and conveniently. All ores are minerals, but all minerals are not ores.

****Gangue (Matrix):**** The unwanted earthly impurities like sand, clay, and rocks associated with the ore.

****2. Common Ores of Iron, Aluminium and Zinc****

****Iron:**** Haematite (Fe_2O_3), Magnetite (Fe_3O_4).

****Aluminium:**** Bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$).

****Zinc:**** Zinc blende (ZnS), Calamine (ZnCO_3).

****3. Stages Involved in the Extraction of Metals****

****A. Pulverisation and Dressing (Concentration) of the Ore****

The ore is crushed into fine powder.

Methods of concentration include:

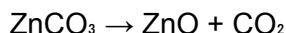
Hydraulic Washing: Based on the difference in densities of ore and gangue particles.

Magnetic Separation: Used when the ore is magnetic and gangue is non-magnetic (or vice versa), e.g., magnetite.

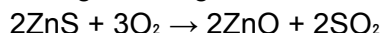
Froth Flotation: Used for sulphide ores. Powdered ore is mixed with water and frothing agents. Air is blown, the ore rises with the froth and is skimmed off.

****B. Conversion of Concentrated Ore to its Oxide****

Calcination: Heating the ore in the absence or limited supply of air. Used for carbonate ores.

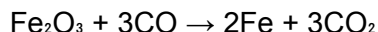
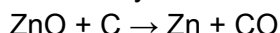


Roasting: Heating the ore in the presence of excess air. Used for sulphide ores.



****C. Reduction of Metallic Oxides to Metal****

Reduction by Carbon/Carbon Monoxide: For less active metals (Zn, Fe, Pb, Cu).



Reduction by Electrolysis: For highly active metals (Na, K, Ca, Mg, Al).

Metals like Al_2O_3 and MgO cannot be reduced by carbon.

****D. Electrefining:** Purification of impure metals by electrolysis.**

****4. Extraction of Aluminium****

****A. Baeyer's Process (Chemical Purification of Bauxite)****

Bauxite ($\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$) contains impurities like Fe_2O_3 and SiO_2 . It is treated with hot concentrated NaOH solution. Aluminium oxide dissolves to form sodium aluminate (NaAlO_2). Iron oxide remains insoluble and is filtered off. The filtrate is cooled, $\text{Al}(\text{OH})_3$ precipitates. $\text{Al}(\text{OH})_3$ is heated strongly to get pure alumina (Al_2O_3).

****B. Hall-Heroult's Process (Electrolytic Reduction)****

Electrolyte: Pure alumina (Al_2O_3) dissolved in molten cryolite (Na_3AlF_6). Cryolite lowers the melting point and improves conductivity.

Electrolytic Cell: Carbon lined steel tank acts as the cathode. Graphite rods act as the anode.

Reaction at Cathode: $2\text{Al}^{3+} + 6\text{e}^- \rightarrow 2\text{Al}$ (Molten aluminium collected at the bottom).

Reaction at Anode: The oxygen produced reacts with graphite anodes to form CO_2 , gradually burning them away.

****5. Alloys and Amalgams****

****Alloy:**** A homogeneous mixture of two or more metals, or a metal and a non-metal, prepared by melting together.

****Amalgam:**** An alloy in which one of the constituent metals is mercury.

****Composition and Uses of Important Alloys:****

****Stainless Steel:**** Fe (Iron) + Cr (Chromium) + Ni (Nickel). Uses: Cutlery, surgical instruments.

****Magnalium:**** Mg (Magnesium) + Al (Aluminium). Uses: Aircraft parts, balance beams.

****Duralumin:**** Al (Aluminium) + Cu (Copper) + Mg (Magnesium) + Mn (Manganese). Uses: Aircraft frames, automobile parts.

****Brass:**** Cu (Copper) + Zn (Zinc). Uses: Electrical fittings, utensils.

****Bronze:**** Cu (Copper) + Sn (Tin). Uses: Statues, medals, ships' propellers.

****Fuse Metal/Solder:**** Pb (Lead) + Sn (Tin). Uses: Soldering electrical wires.

****IMPORTANT QUESTIONS: METALLURGY****

****Q1.**** Define the terms: Mineral, Ore, Gangue.

****Q2.**** Name the most important ore of aluminium and state its formula.

****Q3.**** What is the difference between calcination and roasting? Give one example and equation for each.

****Q4.**** Explain the principle of froth flotation. For which type of ores is it used?

****Q5.**** Describe Baeyer's process for the purification of bauxite. Write relevant equations.

****Q6.**** With a labelled diagram, explain the electrolytic extraction of aluminium by Hall-Heroult's process. Write reactions at the electrodes. Give two reasons why cryolite is added to alumina.

****Q7.**** Why are the graphite anodes replaced periodically in the Hall-Heroult process?

****Q8.**** Give the composition and one use each of:

- (a) Stainless steel
- (b) Brass
- (c) Bronze
- (d) Duralumin

****Q9.**** What is an amalgam? Give one example.

****Q10.**** Arrange the following metals in decreasing order of their reactivity: Cu, Na, Fe, Ag, Al.

****CHAPTER 8: STUDY OF COMPOUNDS – HYDROGEN CHLORIDE (HCl)****

****YouTube Lecture:**** <https://www.youtube.com/live/XnnEcX8IDD0?si=TIIm1QG5vZOKzR2->

****1. Laboratory Preparation of Hydrogen Chloride Gas****

****Reactants:**** Sodium chloride (NaCl) and concentrated Sulphuric acid (H₂SO₄).

****Reaction:**** $\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}\uparrow$ (at low temperature below 200°C)

****Conditions:**** Temperature kept below 200°C.

****Procedure:**** The gas is collected by upward displacement of air (because HCl is heavier than air).

****Precaution:**** The delivery tube should not dip into the water during the fountain experiment/absorption.

****Drying Agent:**** Concentrated sulphuric acid.

****2. Properties of Hydrogen Chloride Gas****

****A. Physical Properties****

Colourless, pungent smelling gas.

Denser than air (V.D. = 18.25, Air = 14.4).

Highly soluble in water - demonstrated by the Fountain Experiment.

****B. Fountain Experiment****

A dry round bottomed flask filled with dry HCl gas is connected to a tube dipping in blue litmus solution. A few drops of water are introduced into the flask. HCl dissolves, creating a partial vacuum. Water rushes in as a red fountain, showing high solubility and acidic nature.

****C. Reaction with Ammonia****

A glass rod dipped in NH_4OH brought near a glass rod dipped in conc. HCl produces dense white fumes of ammonium chloride.

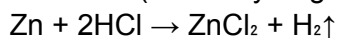


****3. Hydrochloric Acid (Solution of HCl Gas in Water)****

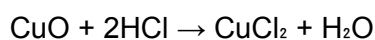
The solution of hydrogen chloride gas in water is called hydrochloric acid. The arrangement for dissolving HCl in water uses a special funnel arrangement to prevent back-suction.

****Acidic Properties of Hydrochloric Acid:****

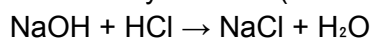
With Metals (above hydrogen in activity series): Forms salt + H_2 .



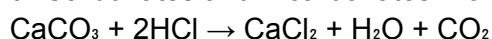
With Metal Oxides: Forms salt + water.



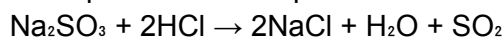
With Metal Hydroxides (Neutralisation): Forms salt + water.



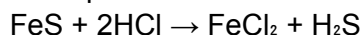
With Carbonates and Bicarbonates: Forms salt + water + CO_2 .



With Sulphites and Bisulphites: Forms salt + water + SO_2 .

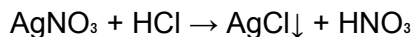


With Sulphides: Forms salt + $\text{H}_2\text{S}\uparrow$.

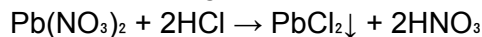


****4. Precipitation Reactions****

Chlorides give a white curdy precipitate of silver chloride with silver nitrate solution (soluble in excess NH_4OH).



With lead nitrate solution, gives a white precipitate of lead chloride (soluble on heating, reappears on cooling).



****5. Aqua Regia****

A mixture of 3 parts concentrated hydrochloric acid and 1 part concentrated nitric acid. It dissolves noble metals like gold and platinum.

****6. Tests for HCl Gas and Hydrochloric Acid****

****HCl Gas:**** Forms dense white fumes with ammonia. Turns blue litmus red.

****Hydrochloric Acid:**** Forms white precipitate with AgNO_3 solution soluble in excess NH_4OH .

****IMPORTANT QUESTIONS: HYDROGEN CHLORIDE****

****Q1.**** With a labelled diagram, describe the laboratory preparation of hydrogen chloride gas. Give the balanced equation.

****Q2.**** Why is HCl collected by upward displacement of air?

****Q3.**** Explain the fountain experiment. Why does a red fountain form when the flask is filled with HCl gas and the water contains blue litmus?

****Q4.**** Write balanced chemical equations for the reaction of hydrochloric acid with:

- (a) Zinc
- (b) Copper oxide
- (c) Sodium hydroxide
- (d) Calcium carbonate
- (e) Sodium sulphite

****Q5.**** What is the name of the funnel used to dissolve HCl gas in water? Why is it used?

****Q6.**** Give two tests to distinguish between HCl gas and hydrochloric acid solution?

****Q7.**** What is aqua regia? State its composition and one use.

****Q8.**** What happens when HCl gas reacts with ammonia gas? Write the equation.

****Q9.**** Name the precipitate formed when silver nitrate solution is added to dilute hydrochloric acid. Is the precipitate soluble in ammonium hydroxide?

****Q10.**** Why is concentrated sulphuric acid not used to dry ammonia gas?

CHAPTER 9: STUDY OF COMPOUNDS – AMMONIA (NH₃)

YouTube Lecture: <https://youtu.be/9DAdgTkVTdg?si=ogLPp6oxMUqAZR XR>

1. Laboratory Preparation of Ammonia Gas

Reactants: Ammonium chloride (NH₄Cl) and Calcium hydroxide [Ca(OH)₂ / slaked lime].

Reaction: $2\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCl}_2 + 2\text{H}_2\text{O} + 2\text{NH}_3\uparrow$

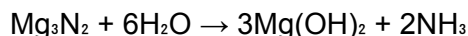
Conditions: Gentle heating.

Drying Agent: Quicklime (CaO). Conc. H₂SO₄ or CaCl₂ cannot be used.

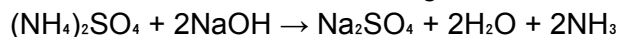
Collection: Downward displacement of air (NH₃ is lighter than air).

2. Preparation of Ammonia from Nitrides and Ammonium Salts

From Nitrides: Action of warm water on magnesium nitride or aluminium nitride.



From Ammonium Salts: Heating ammonium salts with an alkali.



3. Manufacture of Ammonia by Haber's Process

Reactants: Nitrogen and Hydrogen in the ratio 1:3.

Catalyst: Finely divided iron.

Promoter: Molybdenum or alumina.

Temperature: 450 - 500°C.

Pressure: 200 - 900 atmospheres.

Reaction: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 + \text{Heat}$

The process is reversible and exothermic. High pressure favours the forward reaction.

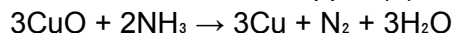
4. Properties of Ammonia Gas

A. Fountain Experiment: Ammonia is highly soluble in water. A flask of dry ammonia creates a vacuum when water is introduced. Water rushes in as a blue fountain (if water contains red litmus) showing its basic nature.

B. Reaction with Hydrogen Chloride: Forms dense white fumes of ammonium chloride.

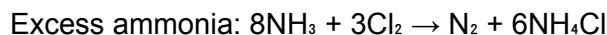


C. Reaction with Hot Copper (II) Oxide: Ammonia reduces CuO to Cu.



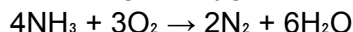
Black copper oxide turns reddish brown.

D. Reaction with Chlorine:



Excess chlorine: $\text{NH}_3 + 3\text{Cl}_2 \rightarrow \text{NCl}_3 + 3\text{HCl}$

****E. Burning in Oxygen:**** Ammonia burns with a greenish-yellow flame.



****5. Aqueous Solution of Ammonia (NH_4OH)****

****Reactions:****

With acids (Neutralisation): $2\text{NH}_4\text{OH} + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4 + 2\text{H}_2\text{O}$

With FeCl_2 solution: Forms dirty green precipitate of $\text{Fe}(\text{OH})_2$.

With CuSO_4 solution: Forms pale blue precipitate of $\text{Cu}(\text{OH})_2$, soluble in excess NH_4OH to form deep blue solution of $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$.

****6. Uses of Ammonia****

Manufacture of fertilizers (ammonium nitrate, ammonium sulphate, urea).

Manufacture of nitric acid (Ostwald's process).

In explosives (ammonium nitrate).

As refrigerant gas (in cold storage).

In cleansing agents (household ammonia).

****7. Tests for Ammonia****

Characteristic pungent smell.

Turns red litmus blue.

Gives dense white fumes with a glass rod dipped in conc. HCl .

With Nessler's reagent, forms a brown precipitate.

****IMPORTANT QUESTIONS: AMMONIA****

****Q1.**** Draw a labelled diagram for the laboratory preparation of ammonia gas. Write the balanced equation. Why is CaO used as the drying agent?

****Q2.**** Explain Haber's process for the manufacture of ammonia with conditions and equation.

****Q3.**** Write the reaction of ammonia with:

(a) Hydrogen chloride gas

(b) Hot copper (II) oxide

(c) Chlorine (in excess)

****Q4.**** Explain the fountain experiment for ammonia. Why is a blue fountain observed?

****Q5.**** Why is ammonium hydroxide called a weak alkali? Write its reaction with sulphuric acid.

****Q6.**** Give two tests to identify ammonia gas.

****Q7.**** State three important uses of ammonia.

****Q8.**** Why can concentrated sulphuric acid not be used as a drying agent for ammonia?

****Q9.**** What happens when ammonia is burnt in excess of oxygen? Write the equation.

****Q10.**** Write the equation for the reaction of ammonia with lead (IV) oxide.

****CHAPTER 10: STUDY OF COMPOUNDS – NITRIC ACID (HNO₃)****

****YouTube Lecture:**** <https://youtu.be/zvirKZpDydg?si=f6WLlxla65K982De>

****1. Laboratory Preparation of Nitric Acid****

****Reactants:**** Potassium nitrate (KNO₃) or Sodium nitrate (NaNO₃) and concentrated Sulphuric acid.

****Reaction:**** $\text{KNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{KHSO}_4 + \text{HNO}_3$ (vapours)

****Temperature:**** Should not exceed 200°C. At higher temperatures, HNO₃ decomposes.

****Apparatus:**** All glass apparatus (retort and receiver).

****Condition:**** Nitric acid vapours are condensed to liquid nitric acid.

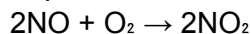
****Colour:**** Pure nitric acid is colourless but the prepared acid is often yellow due to dissolved NO₂.

****2. Manufacture by Ostwald's Process****

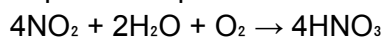
****Step 1: Catalytic oxidation of ammonia.****



****Step 2: Oxidation of nitric oxide to nitrogen dioxide.****



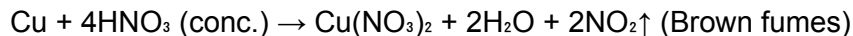
****Step 3: Absorption of NO₂ in water to form nitric acid.****



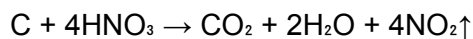
****3. Nitric Acid as an Oxidising Agent****

Concentrated nitric acid is a strong oxidising agent.

****With Copper:****



****With Carbon:****



****With Sulphur:****



****4. Acidic Properties of Dilute Nitric Acid****

With metals (like Mg and Mn, not hydrogen): Magnesium and manganese react with very dilute nitric acid to evolve hydrogen.

With metal oxides/hydroxides/carbonates: Forms corresponding nitrates, water and CO₂ (with carbonates).

Dilute nitric acid is NOT used for preparation of hydrogen from active metals (Zn, Fe) because it is a strong oxidising agent and reduces to NO.

****5. Nitrates and Their Thermal Decomposition****

Nitrates of reactive metals (Na, K) decompose to nitrites and oxygen: $2\text{NaNO}_3 \rightarrow 2\text{NaNO}_2 + \text{O}_2$

Nitrates of less reactive metals (Cu, Pb, Zn) decompose to metal oxide, NO₂ and O₂:
 $2\text{Cu}(\text{NO}_3)_2 \rightarrow 2\text{CuO} + 4\text{NO}_2 + \text{O}_2$

Nitrates of least reactive metals (Ag, Hg) decompose to metal, NO₂ and O₂: $2\text{AgNO}_3 \rightarrow 2\text{Ag} + 2\text{NO}_2 + \text{O}_2$

****6. Tests for Nitric Acid and Nitrates****

****Brown Ring Test:**** To nitrate salt solution, add freshly prepared ferrous sulphate solution. Then carefully add conc. H₂SO₄ along the side of the test tube. A brown ring of ferrous nitrosulphate [Fe(H₂O)₅NO]SO₄ is formed at the junction of the two liquids.

****IMPORTANT QUESTIONS: NITRIC ACID****

****Q1.**** Draw a labelled diagram for the laboratory preparation of nitric acid. Write the balanced equation. Why is all-glass apparatus used?

****Q2.**** Describe Ostwald's process for the manufacture of nitric acid. Give equations.

****Q3.**** Write the reaction of concentrated nitric acid with:

- (a) Copper
- (b) Carbon
- (c) Sulphur

****Q4.**** What happens when copper nitrate is heated strongly? Write the balanced equation.

****Q5.**** Explain the brown ring test for nitrates.

****Q6.**** Why does pure nitric acid turn yellow on standing?

****Q7.**** Why is dilute nitric acid not used to prepare hydrogen from zinc?

****Q8.**** Write equations for the thermal decomposition of sodium nitrate and silver nitrate.

****Q9.**** Why is nitric acid called a strong oxidising agent?

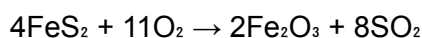
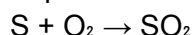
****Q10.**** Why is the temperature kept below 200°C during laboratory preparation of nitric acid?

****CHAPTER 11: STUDY OF COMPOUNDS – SULPHURIC ACID (H₂SO₄)****

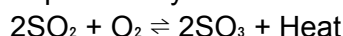
****YouTube Lecture:**** <https://www.youtube.com/live/nVdrzLLRAiY?si=psgRMMJlxFhBCDTU>

****1. Manufacture by Contact Process****

****Step 1: Production of SO₂.** Burning of sulphur or roasting of iron pyrites.



****Step 2: Catalytic oxidation of SO₂ to SO₃.**



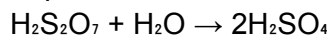
Catalyst: Vanadium pentoxide (V₂O₅) or platinised asbestos.

Temperature: 450°C. Pressure: 1-2 atm.

****Step 3: Absorption of SO₃ in conc. H₂SO₄ to form oleum.**



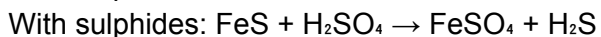
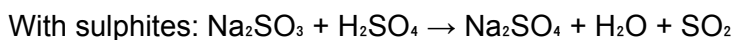
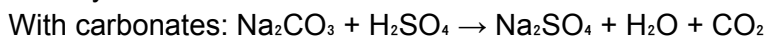
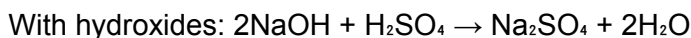
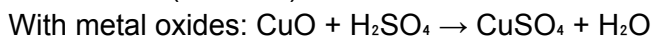
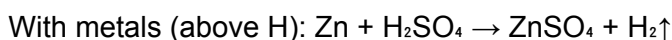
****Step 4: Dilution of oleum with water.**



SO₃ is absorbed in conc. H₂SO₄ and not directly in water because the reaction is highly exothermic and produces a dense fog of acid mist.

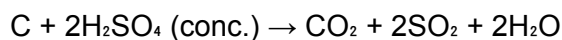
****2. Behaviour as an Acid When Dilute (Dilute H₂SO₄)****

H₂SO₄ is a dibasic acid.

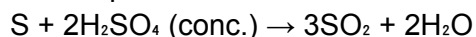


****3. Concentrated Sulphuric Acid as an Oxidising Agent****

****With Carbon:****



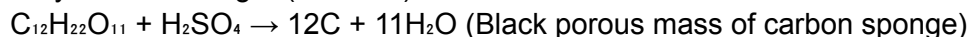
****With Sulphur:****



****4. Concentrated Sulphuric Acid as a Dehydrating Agent****

Conc. H₂SO₄ has a strong affinity for water and removes water from substances.

****Dehydration of Sugar (Sucrose):****



****Dehydration of Copper (II) Sulphate Crystals:****

$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} + \text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$ (Blue crystals turn white, anhydrous copper sulphate)

****5. Non-Volatile Nature of Sulphuric Acid****

Sulphuric acid is non-volatile (has a high boiling point). It can displace more volatile acids from their salts.

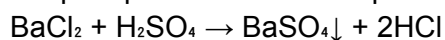
With Sodium Chloride: $\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}\uparrow$

With Sodium Nitrate: $\text{NaNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HNO}_3$

****6. Tests for Sulphuric Acid and Sulphates****

****Aqueous solution of sulphuric acid or sulphate + Barium Chloride:****

White precipitate of barium sulphate (BaSO_4) which is insoluble in all acids (dil. HCl / HNO_3).



****Aqueous solution of sulphuric acid or sulphate + Lead Nitrate:****

White precipitate of lead sulphate (PbSO_4).

****IMPORTANT QUESTIONS: SULPHURIC ACID****

****Q1.**** Explain the Contact Process for the manufacture of sulphuric acid with equations. Why is SO_3 not absorbed directly in water?

****Q2.**** Why is sulphuric acid called a dibasic acid? Give an equation to support your answer.

****Q3.**** Write balanced equations for the reaction of dilute sulphuric acid with:

- (a) Iron
- (b) Zinc carbonate
- (c) Sodium hydroxide

****Q4.**** Explain the action of concentrated sulphuric acid as a dehydrating agent with reference to sugar and copper sulphate crystals.

****Q5.**** What happens when concentrated sulphuric acid is added to sodium chloride? Write the equation. Which property of sulphuric acid is shown here?

****Q6.**** Give a test to distinguish between hydrochloric acid and sulphuric acid.

****Q7.**** Write equations for the reaction of concentrated sulphuric acid with carbon and sulphur. Which property is exhibited?

****Q8.**** What is oleum? Give the equation for its formation.

****Q9.**** Why is vanadium pentoxide preferred over platinised asbestos in the Contact Process?

****Q10.**** A white precipitate insoluble in dilute HCl is formed when an unknown salt solution is treated with barium chloride solution. Identify the anion present.

****CHAPTER 12: ORGANIC CHEMISTRY****

****YouTube Lecture:**** https://youtu.be/_YYL7OBE11k?si=OnSKYj3ssM6L5P2w

****1. Introduction to Organic Compounds****

Organic compounds are compounds of carbon (except CO, CO₂, carbonates and bicarbonates).

****Unique Nature of Carbon:****

Catenation: Carbon atoms can link together to form long chains or rings.

Tetravalency: Carbon has four valence electrons and forms four covalent bonds.

Bond types: Carbon can form single, double, and triple bonds.

****2. Structure and Isomerism****

****Structural Formulae of Hydrocarbons:****

Alkanes: Single bonds only. General formula C_nH_{2n+2}.

Alkenes: Contain a double bond. General formula C_nH_{2n}.

Alkynes: Contain a triple bond. General formula C_nH_{2n-2}.

****Isomerism:**** The phenomenon where two or more organic compounds have the same molecular formula but different structural formulae (arrangement of atoms).

Chain Isomerism: Difference in the arrangement of carbon skeleton.

Position Isomerism: Difference in the position of the functional group or multiple bond.

****3. Homologous Series****

A series of organic compounds where each successive member differs by a CH₂ group.

Characteristics: Same general formula, same functional group, similar chemical properties, gradual gradation in physical properties.

****4. Nomenclature (IUPAC)****

Prefix: Indicates the number of carbon atoms (Meth-, Eth-, Prop-, But-, Pent-).

Suffix: Indicates the type of bond (ane, ene, yne).

Functional group suffix: ol (alcohol), al (aldehyde), one (ketone), oic acid (carboxylic acid).

Simple nomenclature for compounds with one functional group. Follow the longest chain rule and assign the smallest number to the functional group.

****5. Hydrocarbons: Alkanes, Alkenes, Alkynes****

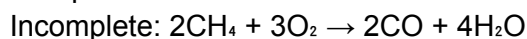
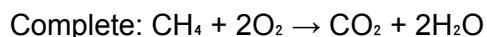
****A. Alkanes (e.g., Methane CH₄)****

Preparation of Methane: Heating sodium ethanoate with soda lime (NaOH + CaO).



Preparation of Ethane: Heating sodium propanoate with soda lime.

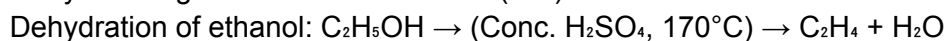
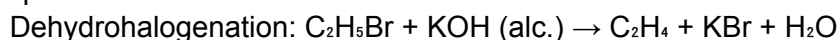
Combustion:



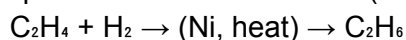
Substitution reaction with chlorine: $\text{CH}_4 + \text{Cl}_2 \xrightarrow{(\text{sunlight})} \text{CH}_3\text{Cl} + \text{HCl}$ (Successive replacements possible)

****B. Alkenes (e.g., Ethene C_2H_4)****

Preparation of Ethene:

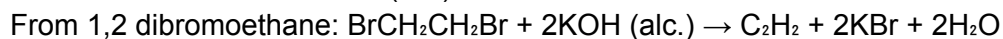
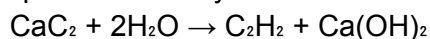


Properties: Addition reactions (unsaturated).

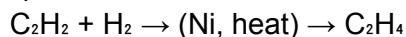


****C. Alkynes (e.g., Ethyne C_2H_2 / Acetylene)****

Preparation of Ethyne: Action of water on calcium carbide.



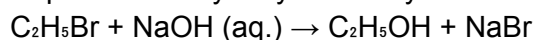
Properties: Addition reactions.



Uses: Oxy-acetylene flame for welding. Manufacture of PVC, synthetic rubber.

****6. Alcohols (Ethanol $\text{C}_2\text{H}_5\text{OH}$)****

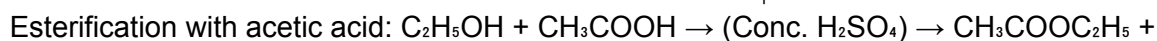
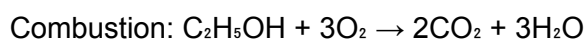
****Preparation:**** Hydrolysis of alkyl halide.



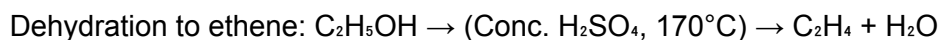
****Properties:****

Physical: Colourless liquid, sweet smell, miscible with water in all proportions.

Chemical:



H_2O



****Denatured Alcohol:**** Ethanol made unfit for drinking by adding poisonous substances.

****Uses:**** Solvent, in medicines, as a fuel, in manufacture of alcoholic beverages.

****7. Carboxylic Acids (Acetic Acid CH_3COOH)****

****Structure:**** Methyl group (CH_3) attached to carboxyl group (COOH).

****Properties:****

Physical: Colourless liquid, pungent vinegar-like smell. On cooling, it freezes to form ice-like crystals called glacial acetic acid.

Chemical:

Weak acid, turns blue litmus red.

Reacts with alkalis: $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

Esterification: Forms esters with alcohols in presence of conc. H_2SO_4 .

Uses: As vinegar (preservative), in manufacture of cellulose acetate, in production of esters.

IMPORTANT QUESTIONS: ORGANIC CHEMISTRY

Q1. Define catenation. Why does carbon show catenation to a large extent?

Q2. What is isomerism? Draw the structural isomers of butane (C_4H_{10}).

Q3. Define homologous series. State any two characteristics of a homologous series.

Q4. Write the IUPAC names of the following:

(a) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH}$

(b) $\text{CH}_3\text{-COOH}$

(c) $\text{CH}_3\text{-CH=CH}_2$

(d) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$

Q5. Describe the laboratory preparation of methane with a diagram and equation.

Q6. How is ethene prepared from ethanol? Write the equation and conditions.

Q7. Write the reactions of ethyne with:

(a) Hydrogen (in presence of Ni)

(b) Bromine water

Q8. Describe the esterification reaction between ethanol and acetic acid with equation and conditions.

Q9. What is glacial acetic acid? Why is it called so?

Q10. Give two differences between alkanes and alkenes.

Q11. What is denatured alcohol?

Q12. Write the general formula of alkanes, alkenes and alkynes.